

Predictive Modeling of Heart Disease Risk Using Health Conditions and Behavioral Factors

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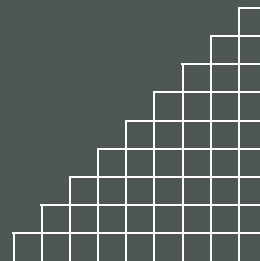


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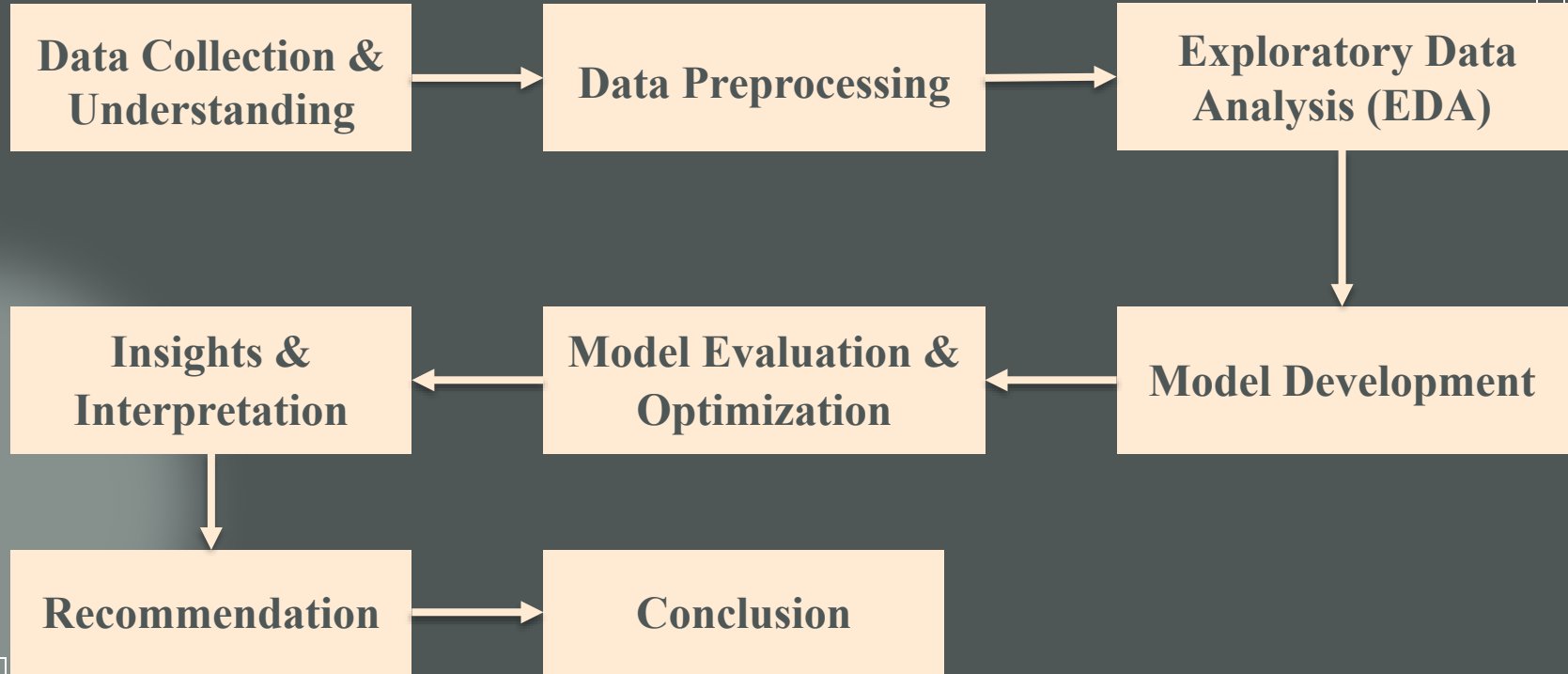
Introduction

- Cardiovascular diseases are the leading cause of death globally, causing nearly 18 million deaths each year.
- Traditional risk scoring methods often overlook individual differences, but applying advanced machine learning techniques to comprehensive health data enables more precise, personalized risk predictions.
- The machine learning approach empowers patients to proactively manage their heart health and strengthens preventative care efforts.

Problem Statement

- This project aims to build a machine learning-based predictive model to assess heart disease risk by analyzing health conditions, behavioral patterns, and mental health factors. By integrating these diverse aspects, the model delivers a more accurate and personalized heart risk assessment, helping enable earlier detection, targeted prevention, and better health outcomes.

Process Flow



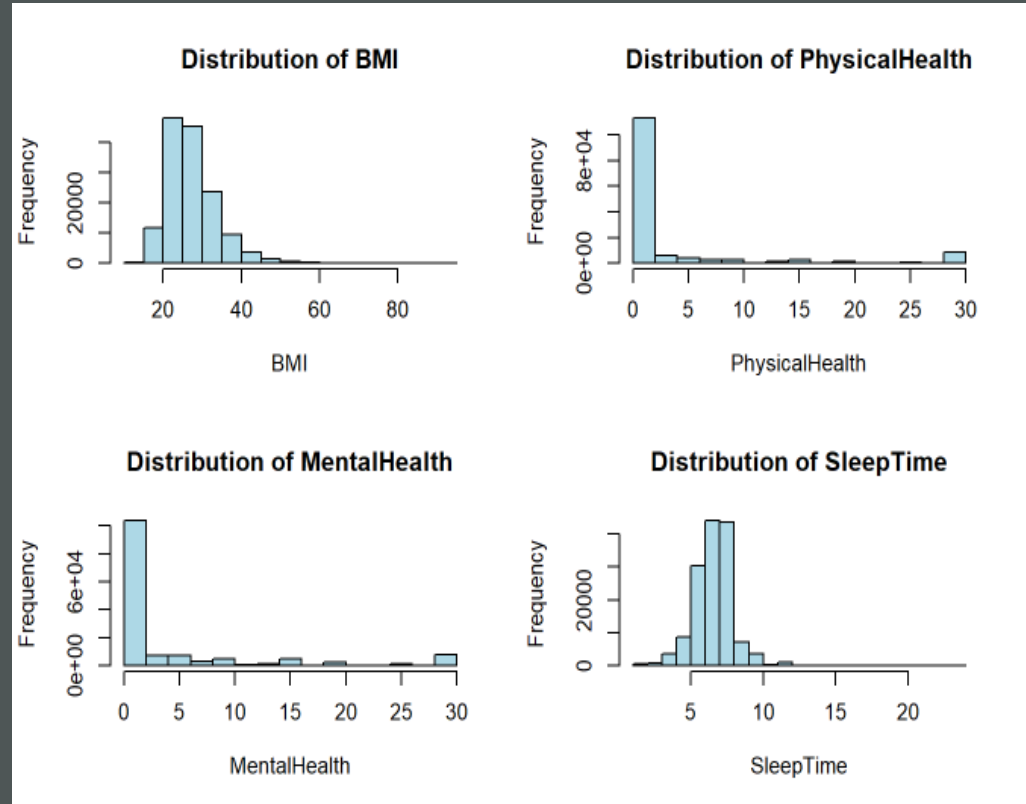
Data Dictionary

- The dataset includes around 140,000 records covering demographic and health-related factors.
- The data was sourced from Kaggle.
- Key features like age, BMI, smoking habits, and physical activity were used, with **Heart Disease** (Yes/No) as the target variable for prediction.
- Preprocessing was required to clean the data and make it ready for the project.

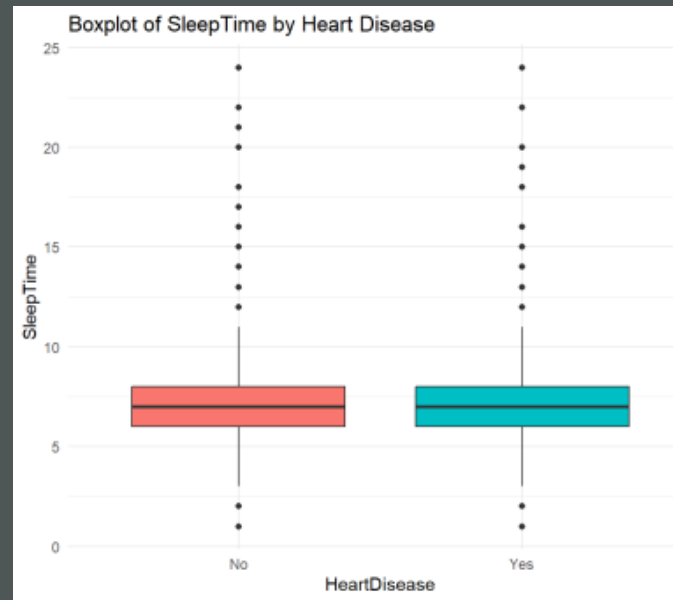
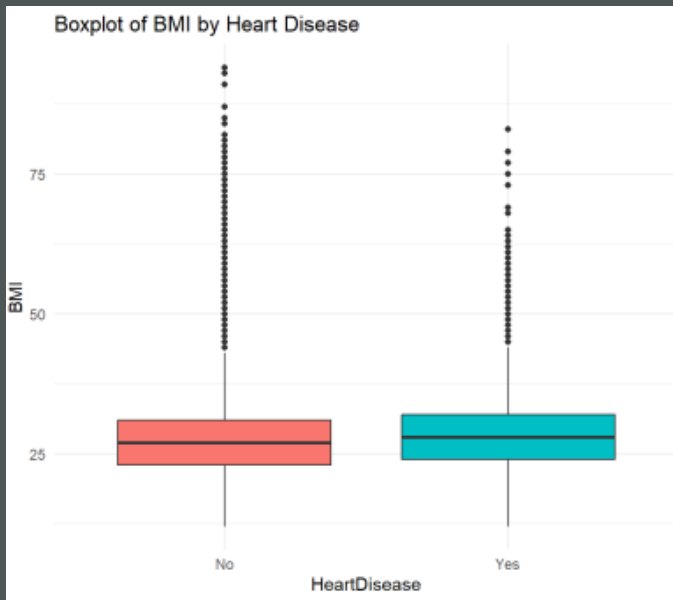
Exploratory Data Analysis

Distribution of numerical variables:

→ The graphs show normal patterns for BMI and sleep, but most people reported very few physical and mental health problems.



Box Plot Graphs for BMI and Sleep-Time



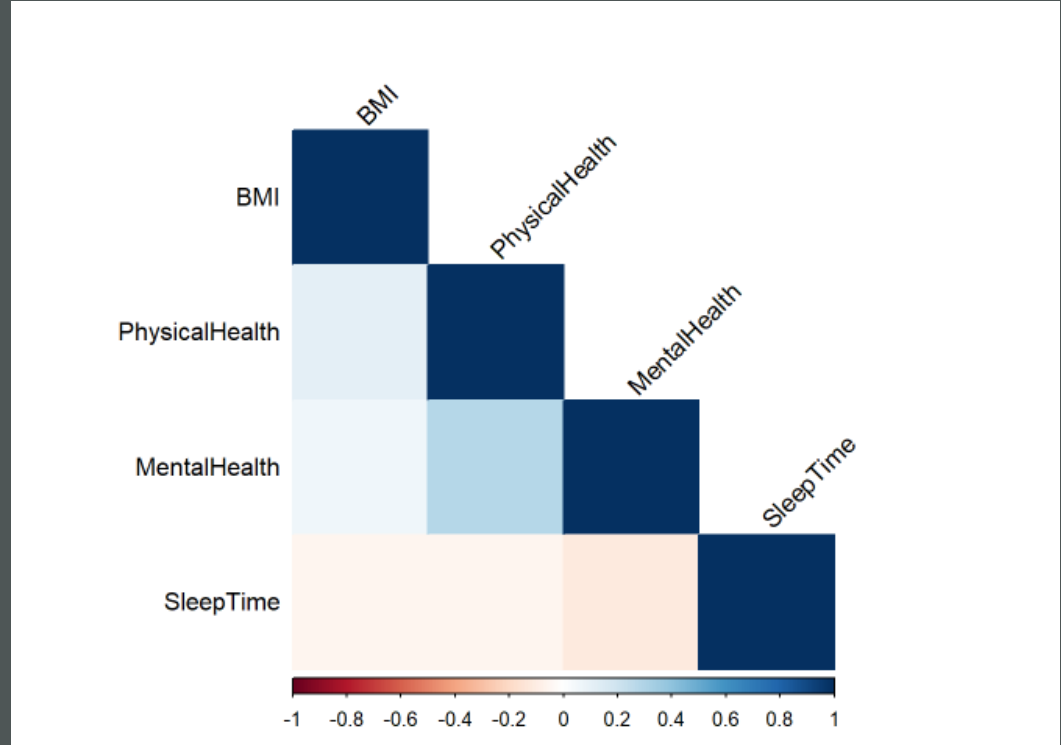
The box plots show slightly higher **BMI** for individuals with heart disease, while **SleepTime** remains similar across both groups around 7–9 hours.

Correlation Heat-Map

The heatmap shows weak to moderate correlations among BMI, Physical Health, Mental Health, and Sleep Time.

→ Physical Health and Mental Health have a slightly stronger positive relationship.

→ Sleep Time shows very low correlation with other variables.



Data Cleaning

- Handled missing values by removing or imputing them appropriately.
- Identified and removed outliers using graphical methods like boxplots and histograms.
- Corrected misclassified entries to ensure consistency across variables.
- Created dummy variables for categorical features to prepare for modeling.
- Explored and verified relationships between variables to check data quality.
- This process prepares the data for advanced analysis, such as regression or classification modeling, to predict heart disease.

Model Development

- Defined Heart Disease as the target variable and other features as predictors.
- Split the data into 80% training set and 20% test set for model evaluation.
- Handled class imbalance using the resampling technique.
- Applied min-max normalization to scale numeric variables uniformly.
- Trained multiple models:
 - Logistic Regression, Decision Tree, Random Forest, XG-Boost, Neural Network, Clustering.
- Evaluated models with metrics like accuracy, sensitivity, specificity, and AUC.

Logistic Regression

Logistic Regression model was built to predict whether an individual has heart disease based on various health and lifestyle-related variables. This model is optimized for binary classification, making it well-suited for distinguishing between individuals with and without heart disease.

It achieved an accuracy of 76.9%,

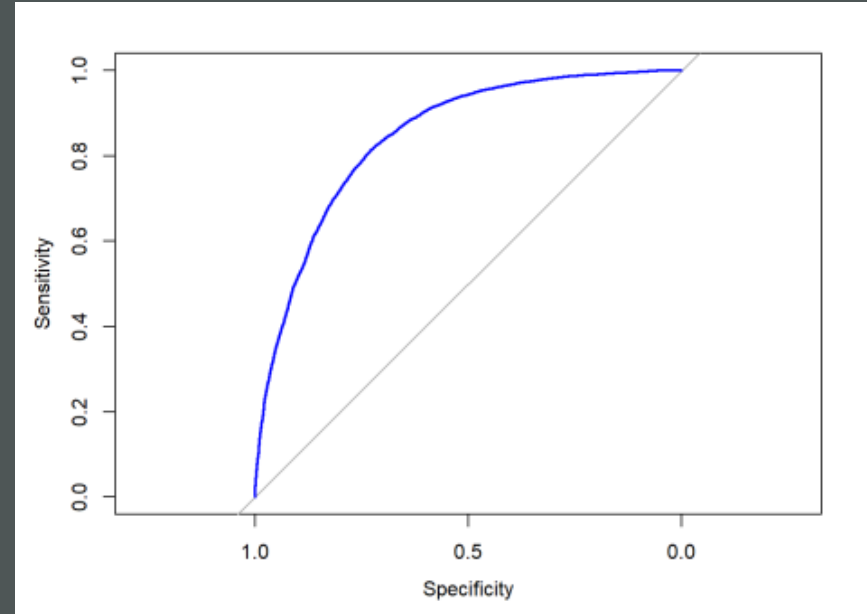
→ Sensitivity: 75.04% – the model was good at detecting people without heart disease.

→ Specificity: 78.69% – also good at identifying those with the condition.

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction   No   Yes
##           No 10440 2947
##           Yes 3472 10880
##
##           Accuracy : 0.7686
##           95% CI : (0.7636, 0.7735)
##           No Information Rate : 0.5015
##           P-Value [Acc > NIR] : < 2.2e-16
##
##           Kappa : 0.5372
##
##           Mcnemar's Test P-Value : 6.14e-11
##
##           Sensitivity : 0.7504
##           Specificity : 0.7869
##           Pos Pred Value : 0.7799
##           Neg Pred Value : 0.7581
##           Prevalence : 0.5015
##           Detection Rate : 0.3764
##           Detection Prevalence : 0.4826
##           Balanced Accuracy : 0.7686
##
##           'Positive' Class : No
##
```

Logistic Regression

The AUC is 0.845, indicating strong classification performance and is quite good and suggests the model has strong discriminatory ability between the two classes. Key variables such as BMI, smoking, diabetes, general health, and age category were found to be significant predictors of heart disease risk, highlighting the importance of both lifestyle and health factors in early detection.



Decision Tree

A Decision Tree model was built to classify individuals as having heart disease or not, based on a set of health-related and behavioral predictors. The model works by learning simple decision rules from the data and is known for being easy to interpret and visualize.

It achieved an accuracy of 74.2%,

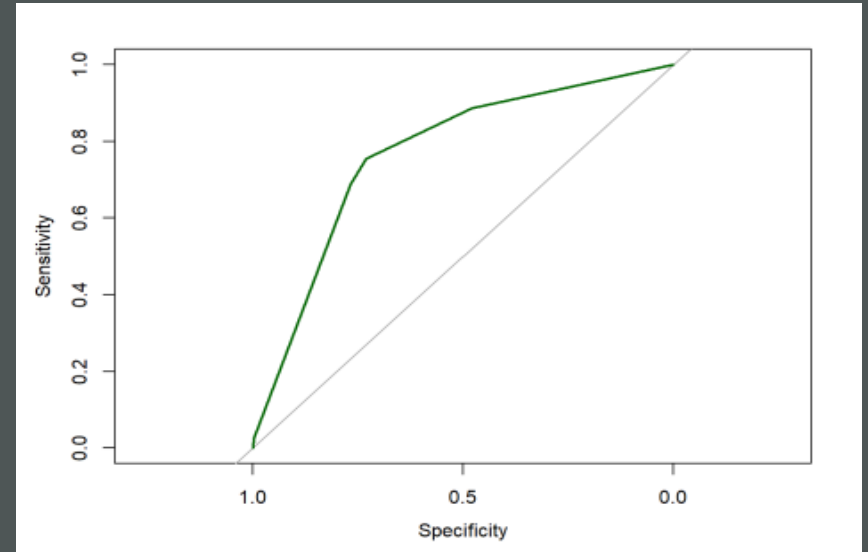
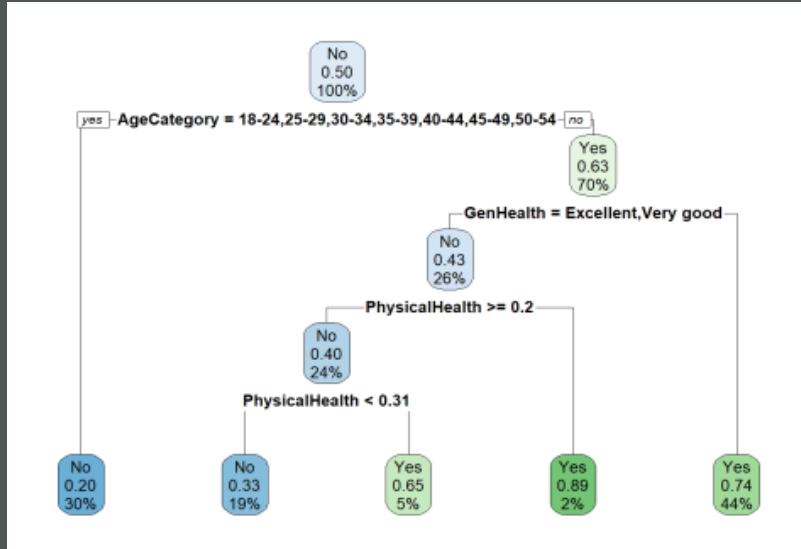
→ Sensitivity: 72.95% – effective at identifying those without heart disease.

→ Specificity: 75.52% – capable of detecting individuals with the condition.

```
## Confusion Matrix and Statistics
##
##               Reference
## Prediction    No    Yes
##          No 10149  3385
##          Yes  3763 10442
##
##               Accuracy : 0.7423
##               95% CI : (0.7371, 0.7475)
##          No Information Rate : 0.5015
##          P-Value [Acc > NIR] : < 2.2e-16
##
##               Kappa : 0.4847
##
## Mcnemar's Test P-Value : 8.23e-06
##
##               Sensitivity : 0.7295
##               Specificity : 0.7552
##          Pos Pred Value : 0.7499
##          Neg Pred Value : 0.7351
##          Prevalence : 0.5015
##          Detection Rate : 0.3659
##          Detection Prevalence : 0.4879
##          Balanced Accuracy : 0.7424
##
##          'Positive' Class : No
##
```

Decision Tree

The AUC is 0.845, indicating strong classification performance and is quite good and suggests the model has strong discriminatory ability between the two classes. Key variables such as BMI, smoking, diabetes, general health, and age category were found to be significant predictors of heart disease risk, highlighting the importance of both lifestyle and health factors in early detection.



Random Forest

The Random Forest model was developed to improve prediction accuracy by combining multiple decision trees and averaging their results. It is known for its robustness and ability to handle complex relationships between variables in large datasets.

It achieved a high accuracy of 83.99%

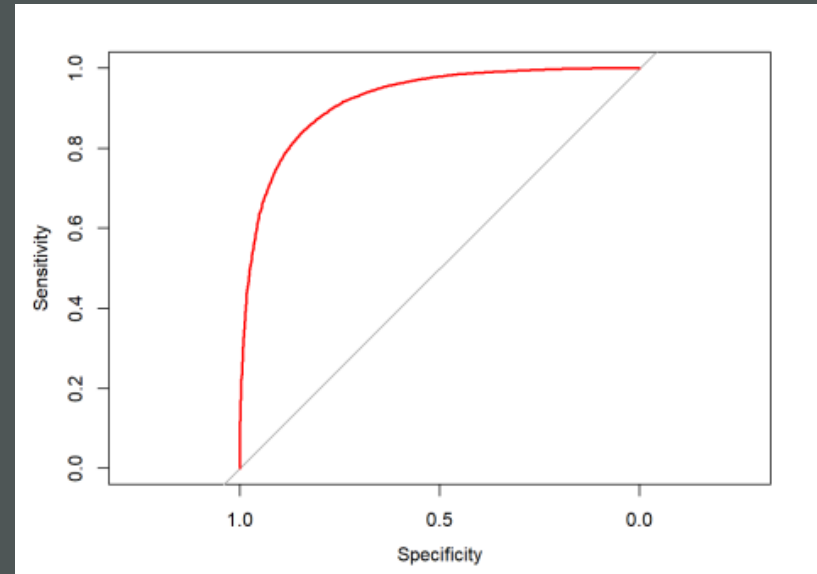
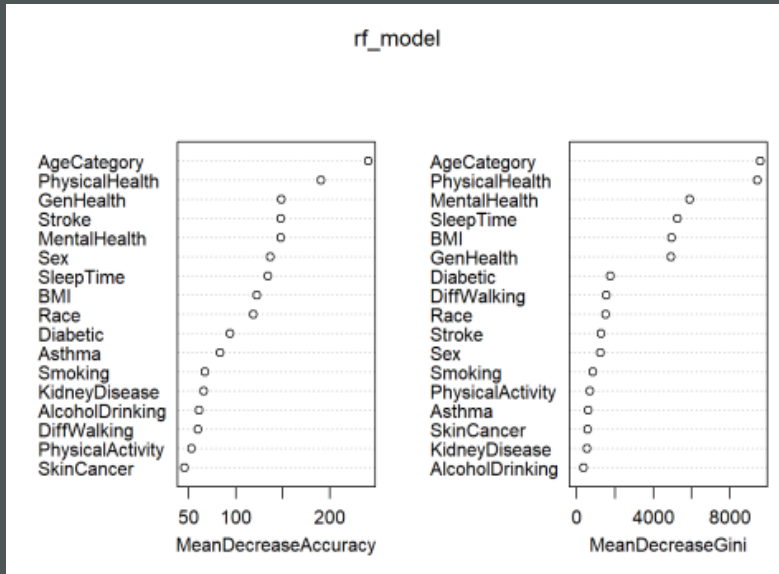
→ Sensitivity: 80.68% – effective at correctly identifying those without heart disease.

→ Specificity: 87.32% – performed very well in detecting individuals with heart disease.

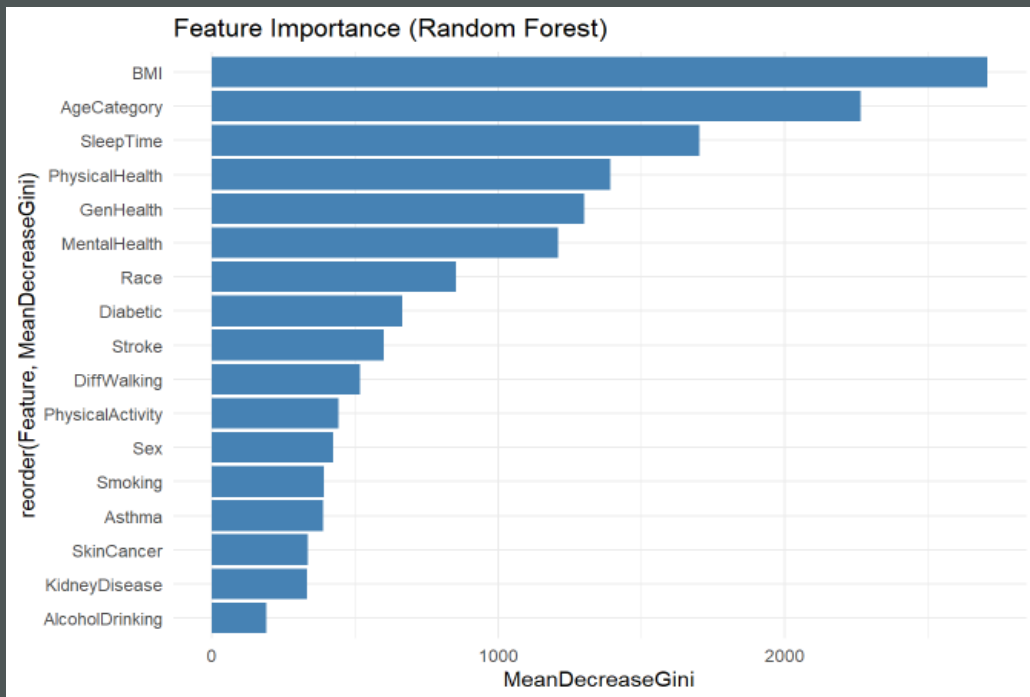
```
## Confusion Matrix and Statistics
##
##               Reference
## Prediction      No      Yes
##           No 11224  1753
##           Yes  2688 12074
##
##               Accuracy : 0.8399
##               95% CI : (0.8355, 0.8442)
##           No Information Rate : 0.5015
##           P-Value [Acc > NIR] : < 2.2e-16
##
##               Kappa : 0.6799
##
## Mcnemar's Test P-Value : < 2.2e-16
##
##           Sensitivity : 0.8068
##           Specificity : 0.8732
##           Pos Pred Value : 0.8649
##           Neg Pred Value : 0.8179
##           Prevalence : 0.5015
##           Detection Rate : 0.4046
##           Detection Prevalence : 0.4678
##           Balanced Accuracy : 0.8400
##
##           'Positive' Class : No
##
```


Random Forest

The ROC curve showed excellent model performance, with an AUC of 0.9211, indicating a strong ability to distinguish between the two classes. , the feature importance values showed that the most influential variables were AgeCategory, PhysicalHealth, MentalHealth, SleepTime, and General Health. This means these factors played a major role in predicting the risk of heart disease.



Feature Importance of Variables (Random Forest)



→ Using the Random Forest model, BMI, Age Category, and Sleep Time were identified as the most important factors for predicting heart disease, while variables like Alcohol Drinking and Kidney Disease showed the least importance.

Neural Networks

A Neural Network model was built to capture complex, non-linear patterns in the data for predicting heart disease. This model mimics the human brain's structure and is capable of learning intricate relationships between health and lifestyle variables.

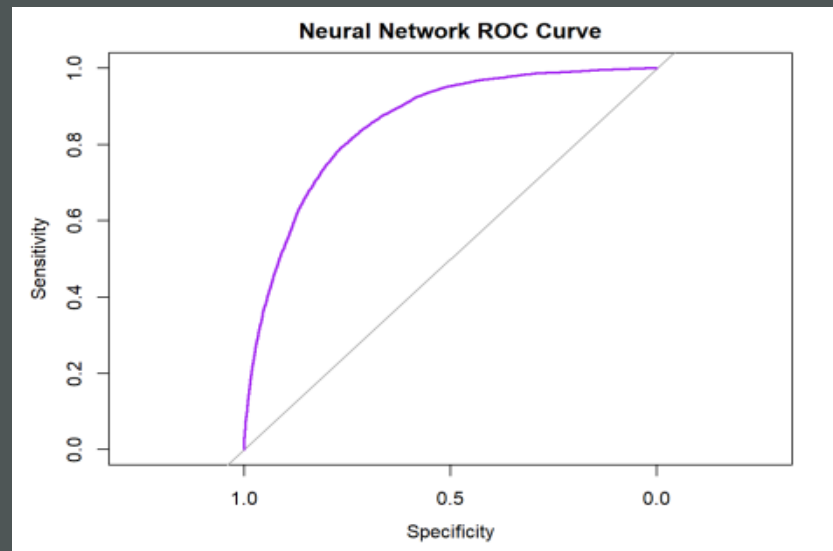
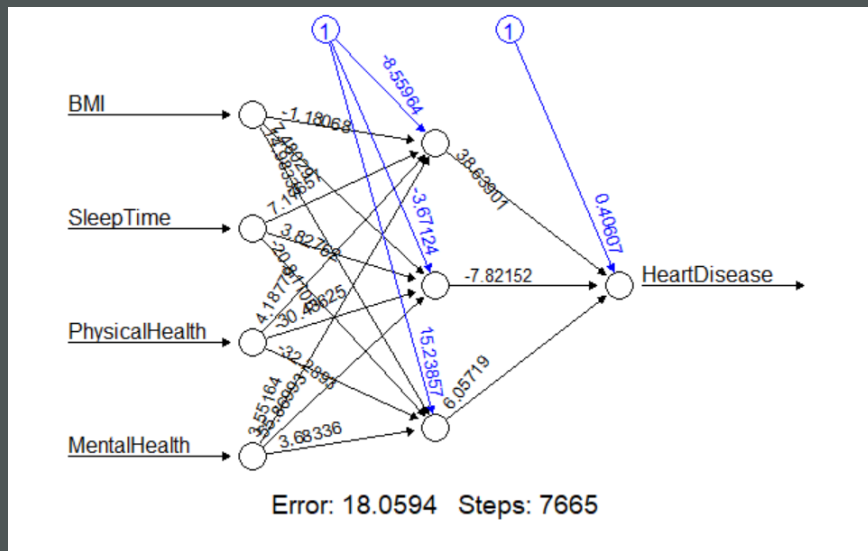
It achieved an accuracy of 77.74%

- Sensitivity: 74.43% — fairly good at identifying individuals without heart disease.
- Specificity: 81.06% — strong in detecting those with the condition.

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction    0    1
##           0 10355 2619
##           1  3557 11208
##
##           Accuracy : 0.7774
##           95% CI : (0.7724, 0.7822)
##           No Information Rate : 0.5015
##           P-Value [Acc > NIR] : < 2.2e-16
##
##           Kappa : 0.5548
##
## Mcnemar's Test P-Value : < 2.2e-16
##
##           Sensitivity : 0.7443
##           Specificity : 0.8106
##           Pos Pred Value : 0.7981
##           Neg Pred Value : 0.7591
##           Prevalence : 0.5015
##           Detection Rate : 0.3733
##           Detection Prevalence : 0.4677
##           Balanced Accuracy : 0.7775
##
##           'Positive' Class : 0
##
```

Neural Networks

Plotted the ROC curve, which showed how the model balances between sensitivity and specificity. The Area Under the Curve (AUC) was 0.8547, indicating a strong ability to distinguish between the two classes. The neural network outperformed several other models in terms of accuracy and AUC, making it a robust choice for heart disease prediction, although it requires more computational resources and is less interpretable compared to simpler models.



XG-Boost Model

The XG-Boost model was implemented to enhance prediction efficiency and accuracy using gradient boosting, which combines multiple weak learners to form a strong predictive model. It is known for its speed, performance, and scalability.

It achieved an accuracy of 79.7%

→ Sensitivity: 79.3% – accurately identified most individuals with heart disease.

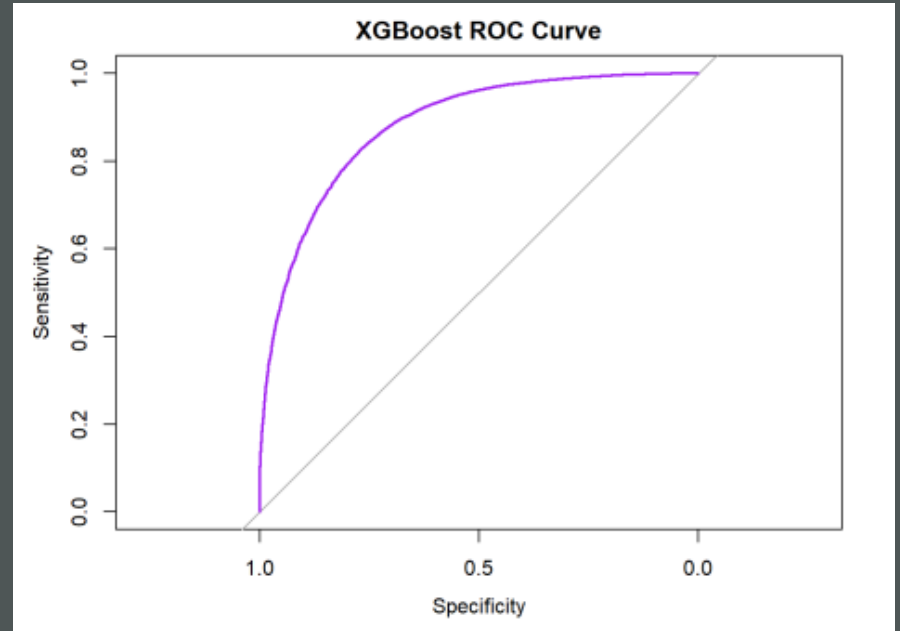
→ Specificity: 80.0% – also reliable in detecting those without the condition.

```
## Confusion Matrix and Statistics
##
##               Reference
## Prediction      0      1
##           0 11037  2761
##           1  2875 11066
##
##               Accuracy : 0.7968
##               95% CI : (0.792, 0.8015)
##               No Information Rate : 0.5015
##               P-Value [Acc > NIR] : <2e-16
##
##               Kappa : 0.5936
##
## Mcnemar's Test P-Value : 0.1323
##
##               Sensitivity : 0.7933
##               Specificity : 0.8003
##               Pos Pred Value : 0.7999
##               Neg Pred Value : 0.7938
##               Prevalence : 0.5015
##               Detection Rate : 0.3979
##               Detection Prevalence : 0.4974
##               Balanced Accuracy : 0.7968
##
##               'Positive' Class : 0
##
```

XG-Boost Model

The AUC (Area Under the Curve) came out to be 0.881, indicating that the model is quite strong.

Overall, XG-Boost outperformed several other models in both accuracy and AUC. It turned out to be a reliable and efficient option for this task. While it's more complex than simpler models like logistic regression, it still runs fairly quickly and doesn't require as much computational power as a neural network, making it a practical choice in many situations.



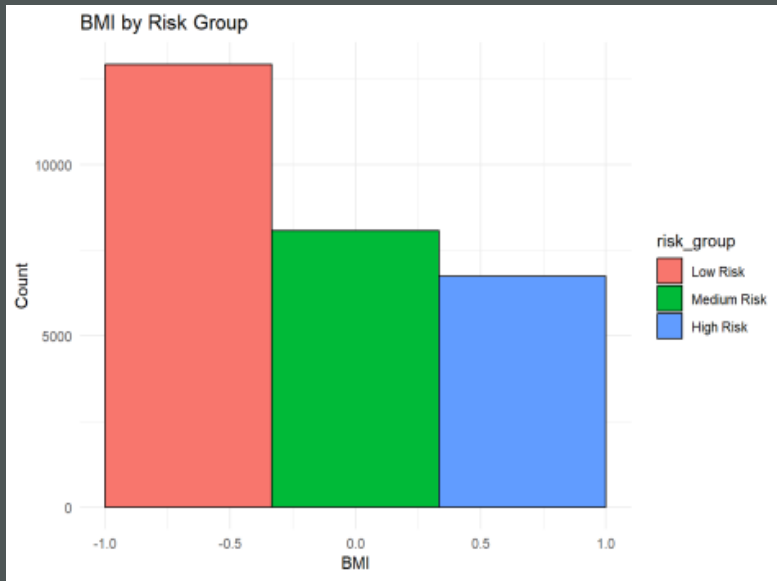
Performance Evaluation

- To compare model performance, we evaluated all five models using accuracy, sensitivity, specificity, and AUC. Among them, Random Forest achieved the highest accuracy (83.9%), sensitivity (80.7%), specificity (87.3%), and AUC (0.921), making it the top performer.
- Although **XG-Boost** and **Neural Networks** also showed strong results, **Random Forest** consistently delivered the best balance across all metrics, proving to be the most effective and reliable model for heart disease prediction in this study.

Regression Models	Sensitivity	Specificity	Accuracy	AUC
Logistic	0.750	0.787	0.769	0.845
Decision Tree	0.729	0.755	0.742	0.767
<u>Random Forest</u>	<u>0.807</u>	<u>0.873</u>	<u>0.839</u>	<u>0.921</u>
Neural Networks	0.744	0.811	0.777	0.855
XG-Boost	0.793	0.800	0.797	0.881

Clustering

- Clustering method is used in data analysis to group similar data points together based on certain features or patterns. The main goal is to make sure that data points within the same group are more alike to each other than to those in other groups.
- Used **Random Forest** to predict probabilities of heart disease.
 - Grouped predicted probabilities into **Low Risk, Medium Risk, and High-Risk** categories.
 - Summarized the data by risk groups, calculating average and standard deviation for numeric features.



BMI Distribution :

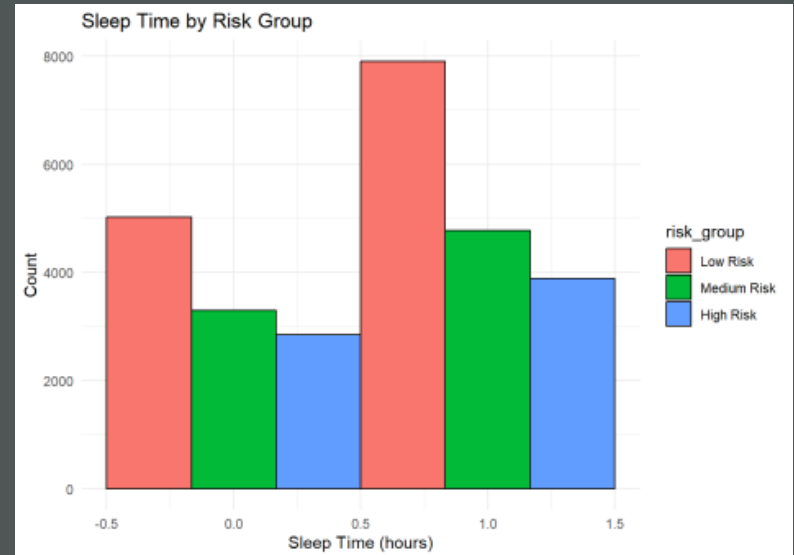
Low-risk group makes up the largest share of the cohort.

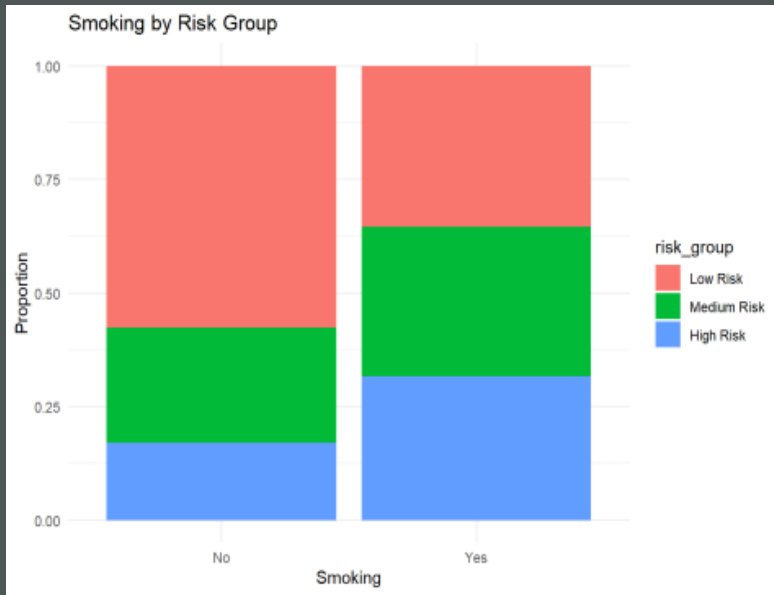
Medium and High-risk groups shrink successively, showing an inverse relationship between BMI-related risk and participant count.

Sleep-Time Distribution :

Low-risk individuals also dominate all sleep-time bands.

Counts drop from **Medium to High risk**, suggesting shorter or irregular sleep is linked to greater health risk.





Smoking and Risk Group Distribution:

The chart shows a clear shift in risk levels based on smoking status:

Non-smokers are mostly in the **Low-Risk** group. Smokers have a higher share of Medium and especially **High-Risk** individuals. This highlights the strong link between smoking and increased health risk.

Age Category vs. Risk Group:

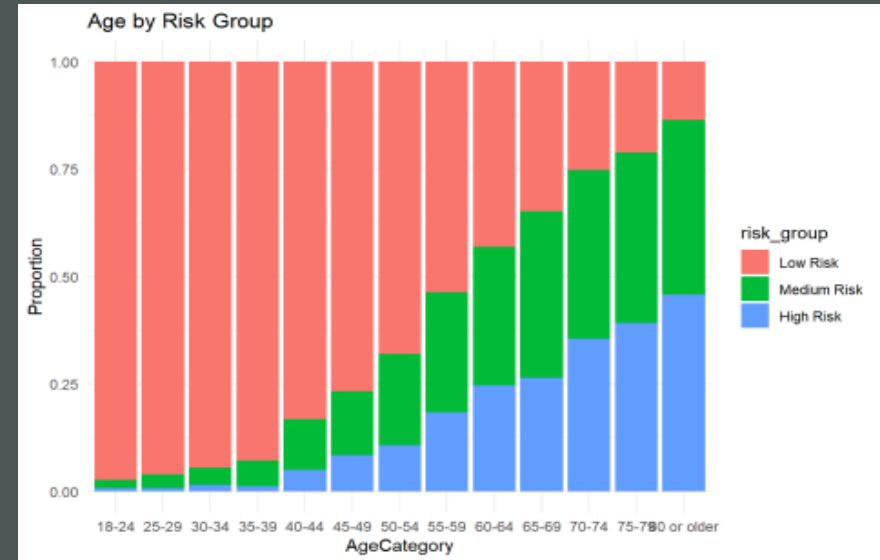
Age clearly influences risk level.

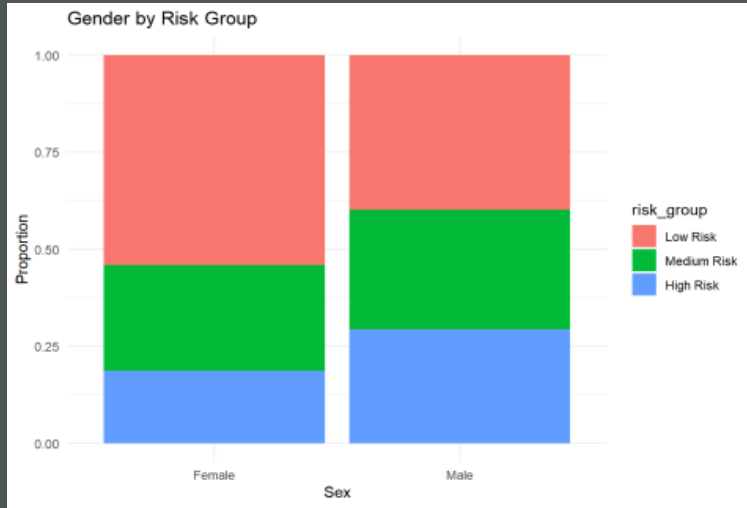
Ages 18–44: Mostly Low Risk.

Ages 45–64: Increasing Medium and High Risk.

Ages 65 and older: Predominantly High Risk.

→ Aging is a strong risk factor. Early lifestyle changes in middle age can reduce future heart disease risk.





Gender vs. Risk Group:

Risk distribution varies slightly by gender.

→ **Females:** Higher proportion in Low-Risk group.

→ **Males:** Higher proportions in Medium and High-Risk groups.

Gender shows a small but noticeable trend, suggesting males may require closer monitoring for heart disease risk.

General Health vs. Risk Group:

Self-rated health strongly aligns with risk level.

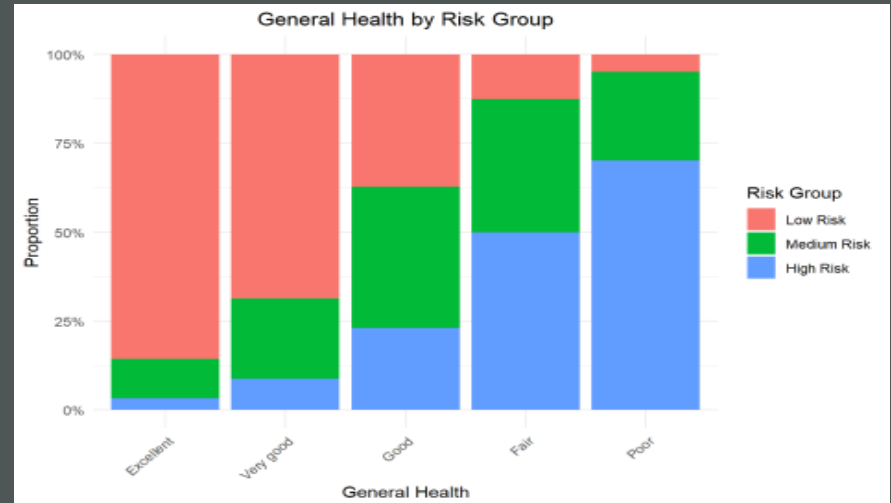
→ **Excellent/Very Good:** Mostly Low Risk

→ **Good:** Higher proportion of Medium Risk

→ **Fair:** Mix of Medium and High Risk

→ **Poor:** Predominantly High Risk

Self-perception of health is an important early indicator for heart disease risk and can guide early interventions.



Recommendation

Personalized Recommendations Based on Risk Categories:

- **High Risk:** Immediate lifestyle changes are needed, intensive exercise plans (after doctor's advice), strict heart-healthy diet, complete smoking and alcohol quitting, regular medical monitoring, and possible medication under supervision.
- **Medium Risk:** Adopt a heart-healthy diet (more fruits, vegetables, low saturated fat), exercise at least 30 minutes daily, and quit smoking or alcohol if applicable.
- **Low Risk:** Maintain current lifestyle. Continue regular check-ups, balanced diet, and physical activity.

Conclusion and Future Work

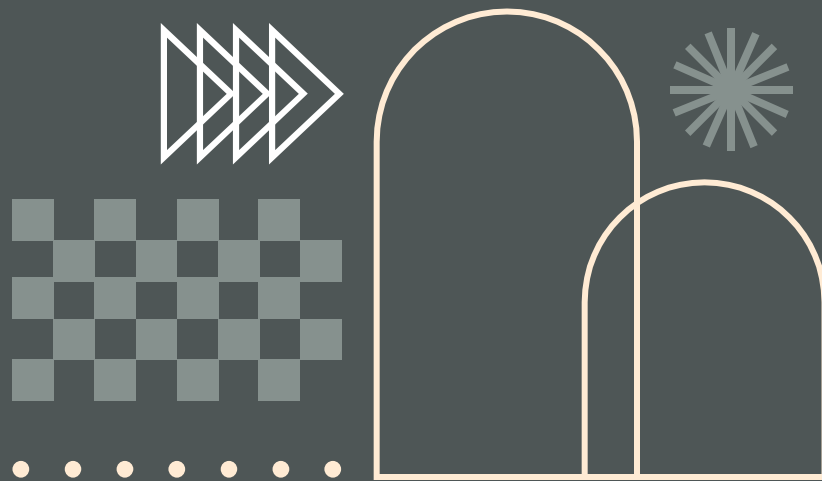
Conclusion

This study developed and compared machine learning models to predict heart disease risk. Random Forest showed the best performance, enabling proactive health management through data-driven strategies.

Future Work

Future research can explore deep learning models, integrate real-time and longitudinal health data, and deploy the model as a clinical decision-support tool for healthcare providers.

Thank You





Appendix

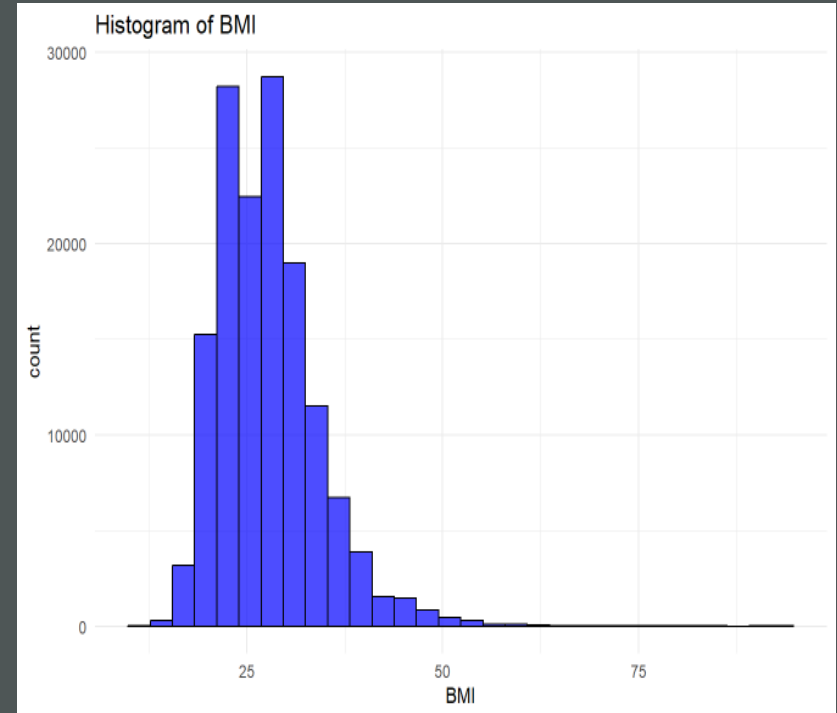
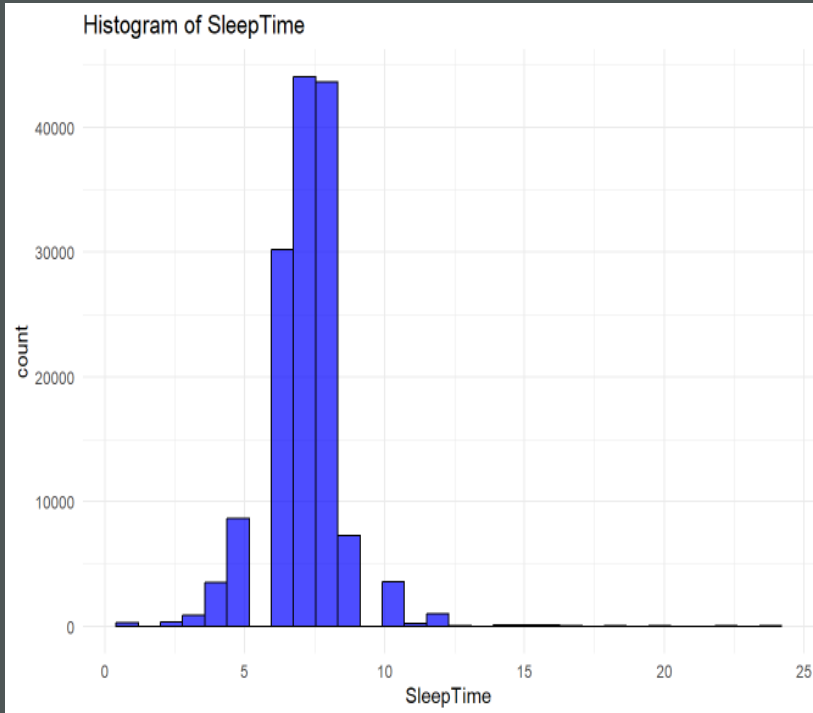


Data Dictionary

Column Name	Description
Heart Disease	Whether the individual has heart disease (Yes/No).
BMI	Body Mass Index (BMI) of the individual.
Smoking	Whether the individual has smoked at least 100 cigarettes in their lifetime (Yes/No).
Alcohol Drinking	Whether the individual is a heavy drinker (Yes/No).
Stroke	Whether the individual has had a stroke (Yes/No).
Physical Health	Number of days in the past 30 days the individual had poor physical health.
Mental Health	Number of days in the past 30 days the individual had poor mental health.
Diff Walking	Whether the individual has difficulty walking or climbing stairs (Yes/No).
Sex	Gender of the individual (Male/Female).
Age Category	Age group of the individual (e.g., 55-59, 80 or older, etc.).
Race	Race of the individual (e.g., White, Black, etc.).
Diabetic	Whether the individual has diabetes (Yes/No/Borderline).
Physical Activity	Whether the individual engages in physical activity other than work (Yes/No).
Gen Health	Self-reported general health status (Excellent, very good, Good, Fair, Poor).
Sleep Time	Average number of hours of sleep per day.
Asthma	Whether the individual has asthma (Yes/No).
Kidney Disease	Whether the individual has kidney disease (Yes/No).

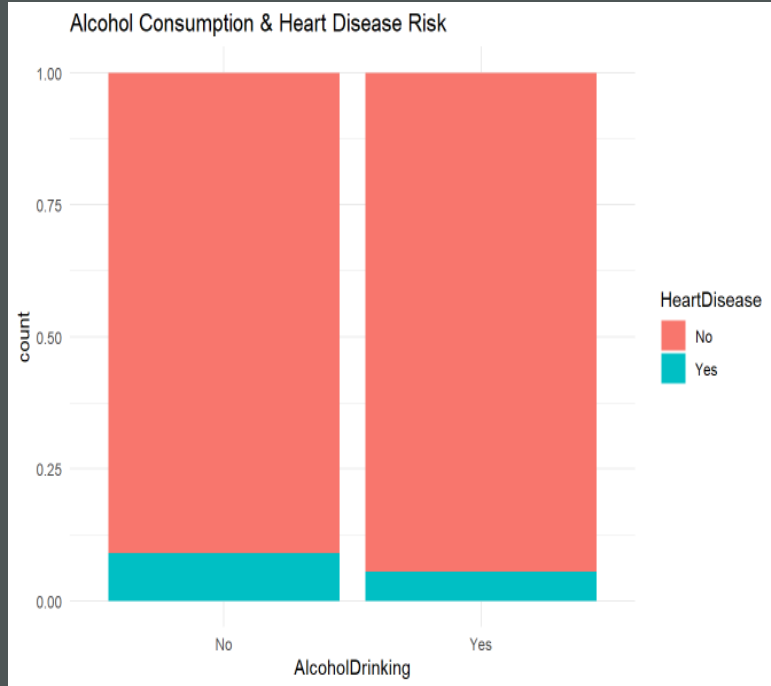
Explanatory Data Analysis

Histogram Graphs for Sleep Time and BMI

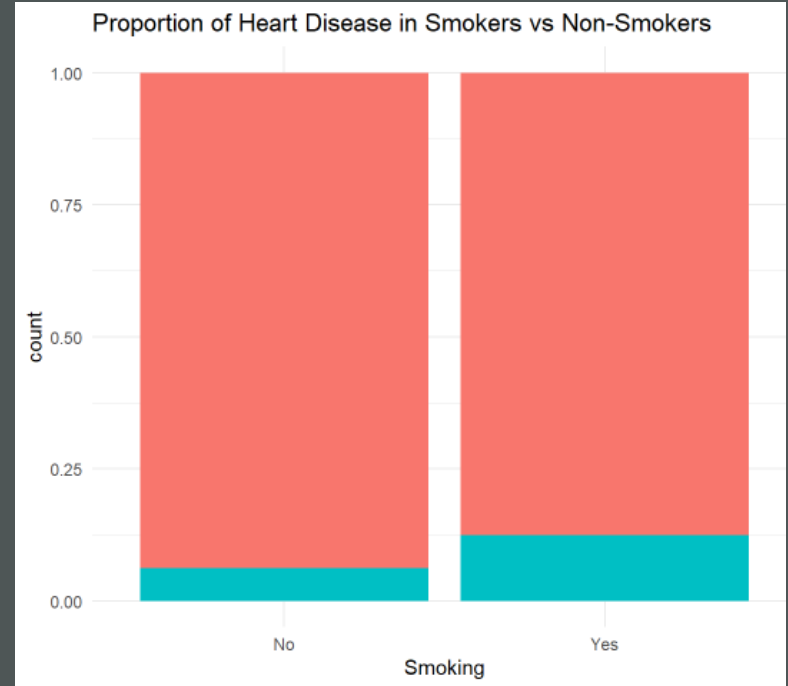


Distribution Graphs

Heart Disease in Alcoholics

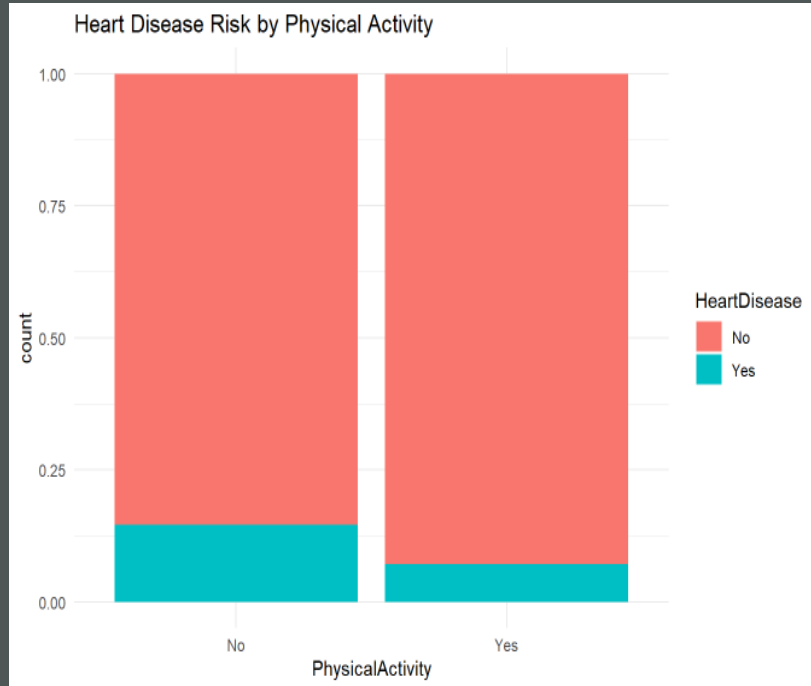


Heart Disease in Smokers and Non-smokers

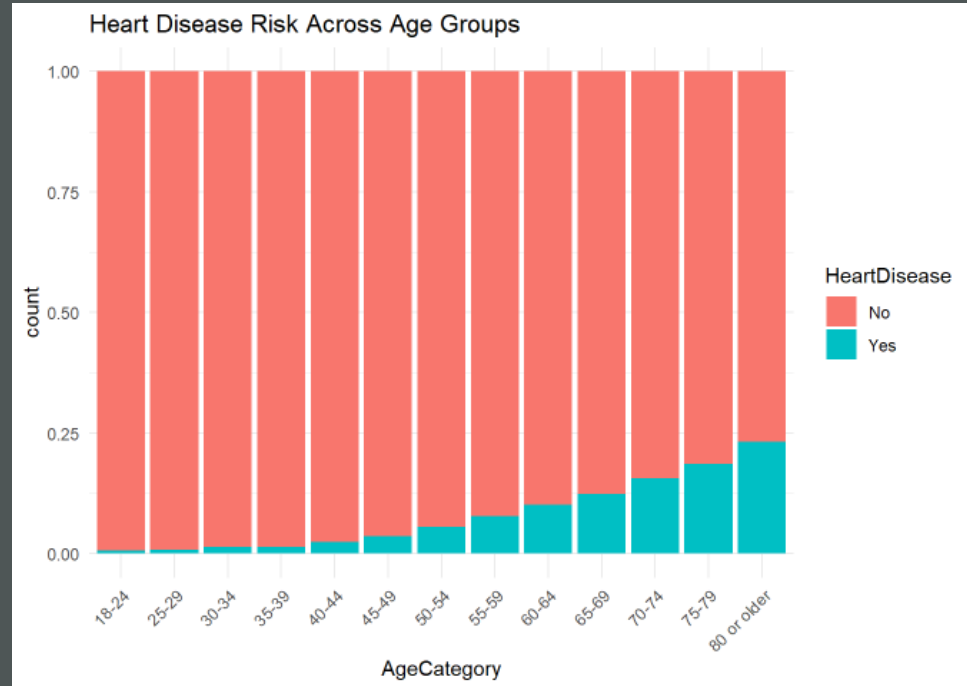


Distribution Graphs

Heart Disease vs Physical health

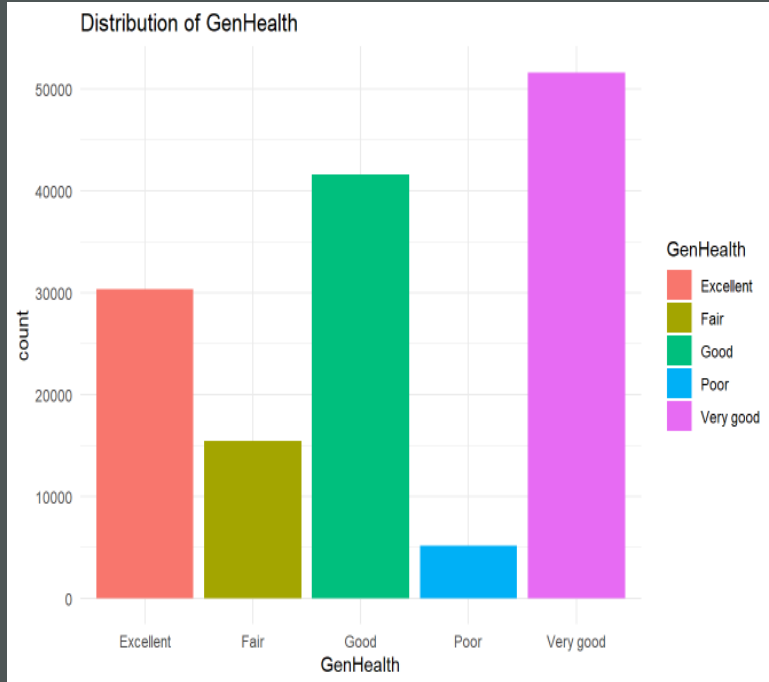


Heart Disease vs Age Category

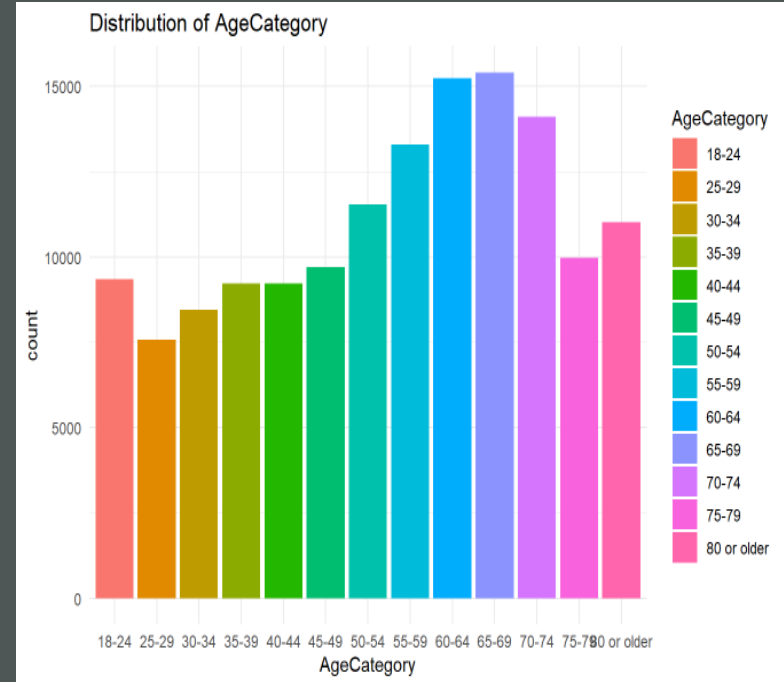


Distribution Graphs

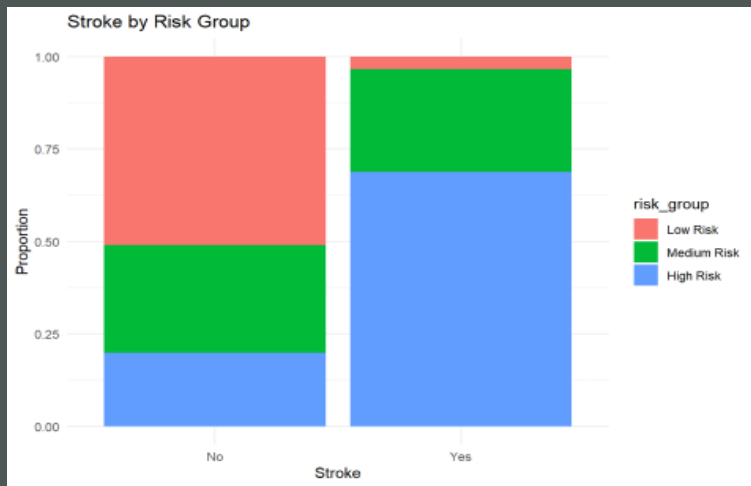
Gen Health



Age Category



Clustering



Diabetic Status vs. Risk Group:

Diabetes status shows a strong impact on heart disease risk.

No Diabetes: Mostly Low Risk

Borderline Diabetes: Higher Medium Risk

Diabetes (Yes): Predominantly High Risk

Gestational Diabetes: Mostly Low Risk

Managing blood sugar levels is crucial to lowering heart disease risk, especially for diagnosed diabetics.

Stroke vs. Risk Group:

A history of stroke is strongly linked to higher heart disease risk.

→ **No Stroke:** Mostly Low and Medium Risk.

→ **Stroke:** Predominantly High Risk.

Stroke history is a major warning sign, highlighting the need for aggressive risk management and monitoring.

